(Please write your Enrolment No. immediately) Enrollment No .....

Mid Term Examination

Second Semester, April ,2024

Paper Code: BS 112 Subject: Applied Mathematics -II

Time $1\frac{1}{2}$ Hrs. Max. Marks:30

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining .

Q		Max. Marks	CO(s)
1(a)	Convert the complex number $\frac{-16}{1+i\sqrt{3}}$ into Polar form .	2.5	1
(b)	Use Cauchy Integral Formula to evaluate $\int_c \frac{\text{Sin}\,\pi z^2 + \text{Cos}\,\pi z^2}{(z-1)(z-2)} dz$,where C is $\|z\|=3$	2.5	1
(c)	Find Invariant Point in $w = \frac{3z-4}{z-1}$	2.5	2
(d)	Determine the residue of function at each Pole. If, $f(z) = \frac{z^2}{(z-1)(z-2)^2}$.	2.5	2
2(a)	Find the Analytic function $f(z) = u + iv$, $Given\ that\ v = e^x(\text{xsin}\,y + y\cos y)$	5	1
(b)	Evaluate the Integral I = $\int_C \text{Re}\,(z^2)dz$ .From Z=0 to 2+4i along the Parabola $y = x^2$	5	1
3(a)	Show that $\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{(a^2-b^2)^{\frac{1}{2}}}$, a> b>0 . Using Contour Integration.	5	2
(b)	Find the Laurent's expansion of $f(z) = \frac{1}{z(z^2-3z+2)}$ For the region $1 \leq \|z\| < 2$.	5	2
4(a)	Find the image of circle $x^2 + y^2 = 4y$ $Under\ the\ transformation\ w = \frac{1}{z}$.	5	2
(b)	Discuss the analyticity of the function $f(z) = z\bar{z}$ at origin.	5	1